Fitbuddy AI Document Blueprint

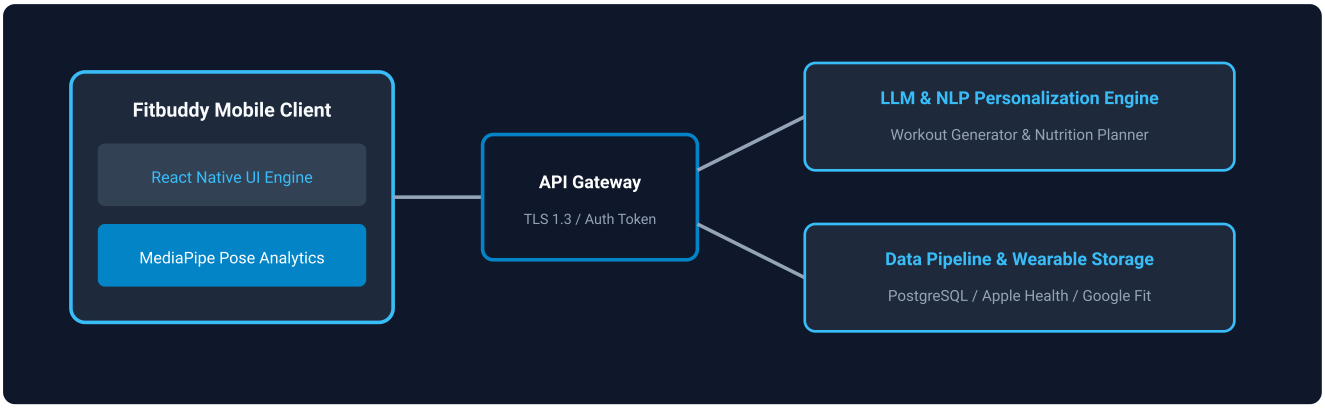
System Architecture, FSD Framework & Final Project Report

PROJECT VERSION	DATE	AUTHOR	TARGET AUDIENCE
v1.0 Final	**September 2026**	**Engineering Team**	**PMs, Devs, Lead QA**

Executive Summary: Fitbuddy AI is an end-to-end fitness companion combining edge-based Computer Vision (CV) pose tracking with Large Language Model (LLM) dynamic meal and workout orchestrations. This document outlines the integrated Functional System Design (FSD) and Final Technical Delivery Report.

1. High-Level Architecture & Data Flow

The system decouples edge pose analytics from cloud-based LLM planning, delivering fast latency for form correction alongside rich personalization.



Edge Pose Processing

MediaPipe runs on-device via web/mobile native assemblies, ensuring 30 FPS rep tracking with minimal server bandwidth cost.

Cloud Orchestration

LLMs process biometrics and goal vectors asynchronously, returning tailored workout and macronutrient matrices.

2. Functional Requirements Document (FRD)

The core application capabilities are cataloged below with key business rules and operational logic:

Module ID	Feature Name	Functional Description & Business Rules	Priority
FR-01	User Onboarding	Captures age, baseline metrics, dietary preferences/allergens, physical limits, and fitness goals.	High
FR-02	Dynamic Workouts	Generates adaptive training routines adjusted for equipment availability and fatigue levels.	High
FR-03	Pose & Form Tracking	Processes device video stream to track joint angles, count completed repetitions, and alert form breaks.	Critical
FR-04	Nutrition Planner	Outputs caloric targets and micronutrient breakdowns while honoring strict allergen exclusions.	Medium
FR-05	Wearable Data Sync	Ingests step counts, resting HR, and active energy burn via HealthKit and Google Fit APIs.	Medium

3. Software Testing & Quality Assurance

A multi-layer QA strategy was deployed covering functional accuracy, edge model performance, and load capacity.

99.2%	30 FPS	< 1.1s
ALLERGEN SAFETY PASS	POSE TRACKING RATE	LLM RESPONSE LATENCY

3.1 Key Validation Focus Areas

- **Basal Metabolic Rate & TDEE Calculation:** Verified math accuracy across 150 diverse test user profiles using Mifflin-St Jeor benchmarks.
- **Rep Counting Accuracy:** Benchmarked MediaPipe joint vectors against standard, fast, and bad-form exercise variations.
- **Allergen Security Protocol:** Automated suite executing 1,000 continuous queries ensuring zero accidental dairy/peanut recommendation leaks.

4. User Acceptance Testing (UAT) & Remediation

Final acceptance testing was executed in September 2026. Results confirmed operational readiness with targeted edge optimization.

Test Category	Executed	Passed	Failed	Outcome Status
User Profile & Onboarding	15	15	0	Approved
AI Workout Generator	22	21	1	Resolved
Real-time Pose Estimation	18	16	2	Remediated
Nutrition & Meal Plans	12	12	0	Approved

UAT Issue & Remediation: Lower-spec hardware experienced frame drops during high-speed pose estimation. Fixed via dynamic resolution scaling that maintains 30 FPS while preserving rep accuracy.

5. Business Intelligence & Deployment Roadmap

BI & Product Analytics

- Track Daily Active Users (DAU) and 30-day program retention.
- Monitor API costs per active user and token utilization rates.
- Measure workout completion rates and form safety indexes.

Next Phase Enhancements

- Multi-person pose tracking for group workout sessions.
- Advanced biometric sync with real-time strain score adjustments.
- Generative audio coaching with natural voice feedback.

Document sign-off complete. Approved for final release & deployment.